

PRACTICAL 3

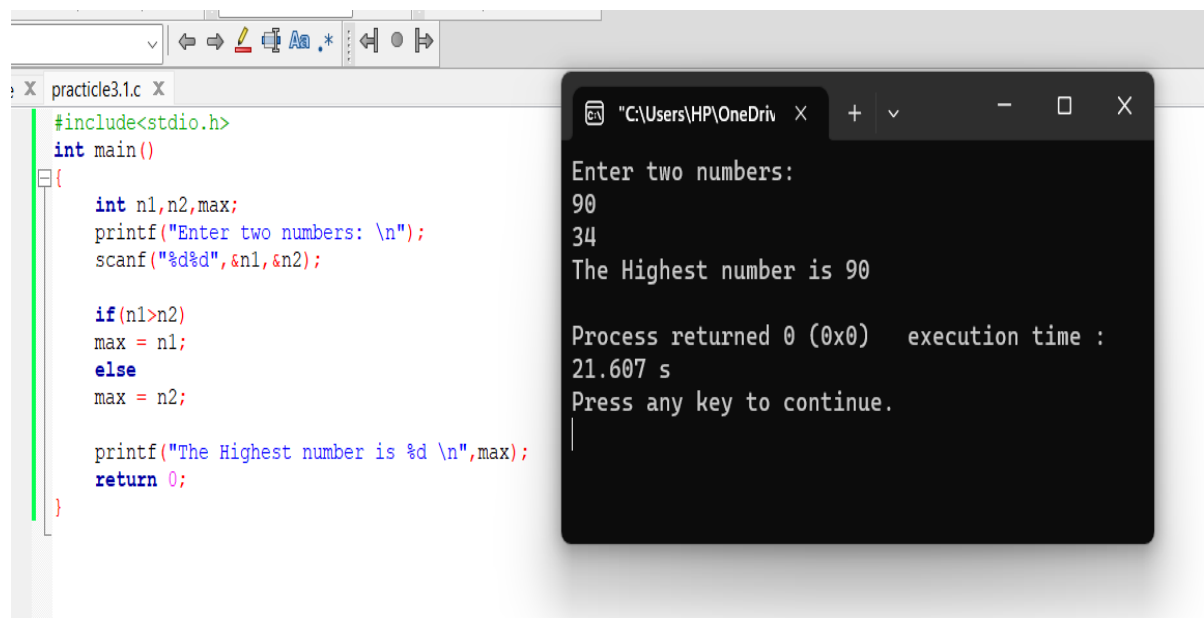
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Batch No : 23.1
Degree Program Name : Technology Management

1.

```
#include<stdio.h>
int main()
{
    int n1,n2,max;
    printf("Enter two numbers: \n");
    scanf("%d%d",&n1,&n2);

    if(n1>n2)
    max = n1;
    else
    max = n2;

    printf("The Highest number is %d \n",max);
    return 0;
}
```



The screenshot displays a code editor window titled 'practicle3.1.c' containing the C program code. To the right, a terminal window shows the execution of the program. The terminal prompts for two numbers, '90' and '34', and outputs 'The Highest number is 90'. It also shows the process returning 0 (0x0) with an execution time of 21.607 s, and prompts the user to press any key to continue.

```
#include<stdio.h>
int main()
{
    int n1,n2,max;
    printf("Enter two numbers: \n");
    scanf("%d%d",&n1,&n2);

    if(n1>n2)
    max = n1;
    else
    max = n2;

    printf("The Highest number is %d \n",max);
    return 0;
}
```

Enter two numbers:
90
34
The Highest number is 90
Process returned 0 (0x0) execution time :
21.607 s
Press any key to continue.

2.

```
#include <stdio.h>
int main()
{
    int n1,n2,n3,largeno,smallno;
    printf("Enter three numbers: ");
    scanf("%d%d%d",&n1,&n2,&n3);

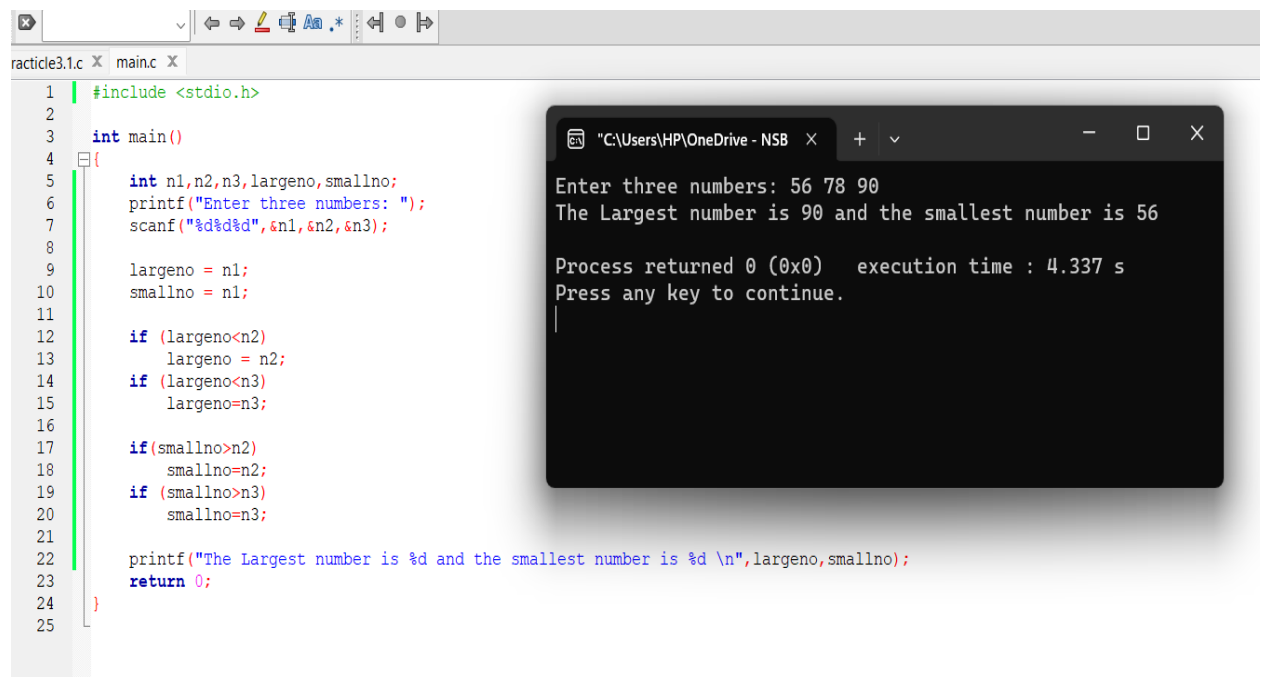
    largeno = n1;
    smallno = n1;

    if (largeno<n2)
        largeno = n2;
    if (largeno<n3)
        largeno=n3;

    if(smallno>n2)
        smallno=n2;
    if (smallno>n3)
        smallno=n3;

    printf("The Largest number is %d and the smallest number is
           %d \n",largeno,smallno);

    return 0;
}
```



```
1 | #include <stdio.h>
2 |
3 | int main()
4 | {
5 |     int n1,n2,n3,largeno,smallno;
6 |     printf("Enter three numbers: ");
7 |     scanf("%d%d%d",&n1,&n2,&n3);
8 |
9 |     largeno = n1;
10 |    smallno = n1;
11 |
12 |    if (largeno<n2)
13 |        largeno = n2;
14 |    if (largeno<n3)
15 |        largeno=n3;
16 |
17 |    if(smallno>n2)
18 |        smallno=n2;
19 |    if (smallno>n3)
20 |        smallno=n3;
21 |
22 |    printf("The Largest number is %d and the smallest number is %d \n",largeno,smallno);
23 |    return 0;
24 | }
25 |
```

```
"C:\Users\HP\OneDrive - NSB" x + v - □ x
Enter three numbers: 56 78 90
The Largest number is 90 and the smallest number is 56

Process returned 0 (0x0) execution time : 4.337 s
Press any key to continue.
```

3.

```
#include <stdio.h>
int main()
{
    char emname[25];
    float basicsalary,newsalary,incre;

    printf("Enter Employee name:");
    scanf("%s",&emname);

    printf("Enter Basic Salary:");
    scanf("%f",&basicsalary);

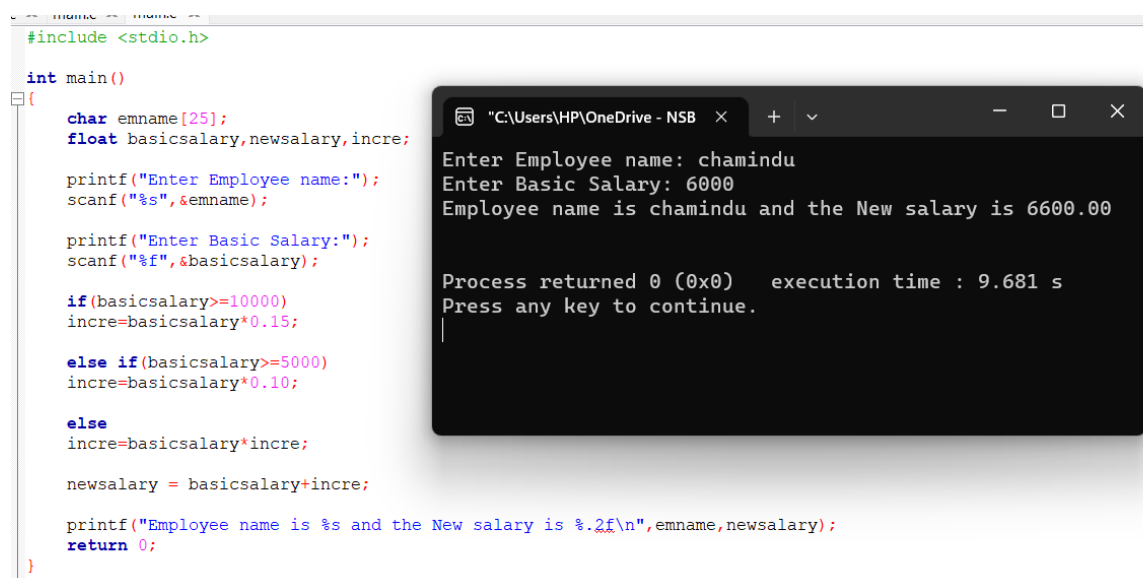
    if(basicsalary>=10000)
        incre=basicsalary*0.15;

    else if(basicsalary>=5000)
        incre=basicsalary*0.10;

    else
        incre=basicsalary*0.05;

    newsalary = basicsalary+incre;

    printf("Employee name is %s and the New salary is
           %.2f\n",emname,newsalary);
    return 0;
}
```



The image shows a code editor on the left and a terminal window on the right. The code editor displays the C program code with syntax highlighting. The terminal window shows the program's execution with the following output:

```
#include <stdio.h>
int main()
{
    char emname[25];
    float basicsalary,newsalary,incre;

    printf("Enter Employee name:");
    scanf("%s",&emname);

    printf("Enter Basic Salary:");
    scanf("%f",&basicsalary);

    if(basicsalary>=10000)
        incre=basicsalary*0.15;

    else if(basicsalary>=5000)
        incre=basicsalary*0.10;

    else
        incre=basicsalary*0.05;

    newsalary = basicsalary+incre;

    printf("Employee name is %s and the New salary is %.2f\n",emname,newsalary);
    return 0;
}
```

```
Enter Employee name: chamindu
Enter Basic Salary: 6000
Employee name is chamindu and the New salary is 6600.00

Process returned 0 (0x0)   execution time : 9.681 s
Press any key to continue.
```

4.

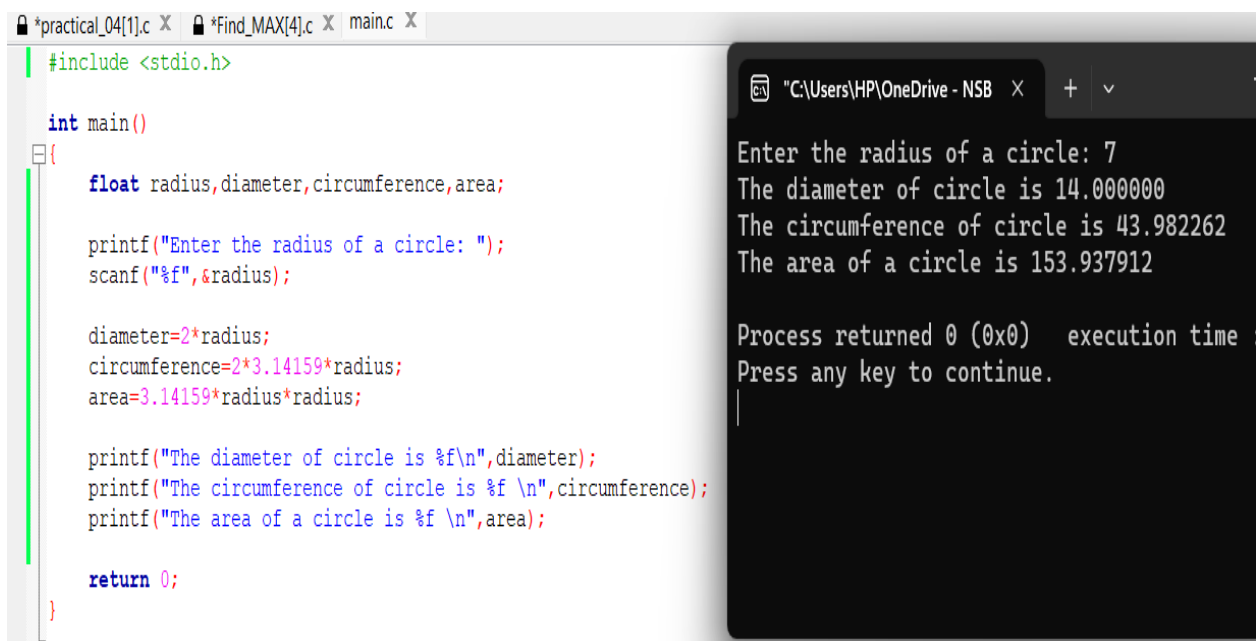
```
#include <stdio.h>
int main()
{
    float radius,diameter,circumference,area;

    printf("Enter the radius of a circle: ");
    scanf("%f",&radius);

    diameter=2*radius;
    circumference=2*3.14159*radius;
    area=3.14159*radius*radius;

    printf("The diameter of circle is %f\n",diameter);
    printf("The circumference of circle is %f\n",circumference);
    printf("The area of a circle is %f \n",area);

    return 0;
}
```



The screenshot shows a C program being executed. On the left, a text editor window displays the source code for a program that calculates the diameter, circumference, and area of a circle based on a user-provided radius. The code includes standard headers, declares floating-point variables, prompts for input, performs calculations using the constant 3.14159, and prints the results. On the right, a console window shows the program's execution. It prompts the user to enter a radius, which is 7. The program then outputs the calculated diameter (14.000000), circumference (43.982262), and area (153.937912). Finally, it displays the message 'Process returned 0 (0x0) execution time' and 'Press any key to continue.'

```
*practical_04[1].c X *Find_MAX[4].c X main.c X
#include <stdio.h>
int main()
{
    float radius,diameter,circumference,area;

    printf("Enter the radius of a circle: ");
    scanf("%f",&radius);

    diameter=2*radius;
    circumference=2*3.14159*radius;
    area=3.14159*radius*radius;

    printf("The diameter of circle is %f\n",diameter);
    printf("The circumference of circle is %f \n",circumference);
    printf("The area of a circle is %f \n",area);

    return 0;
}
```

"C:\Users\HP\OneDrive - NSB X + v

Enter the radius of a circle: 7
The diameter of circle is 14.000000
The circumference of circle is 43.982262
The area of a circle is 153.937912

Process returned 0 (0x0) execution time
Press any key to continue.

5. `#include <stdio.h>`

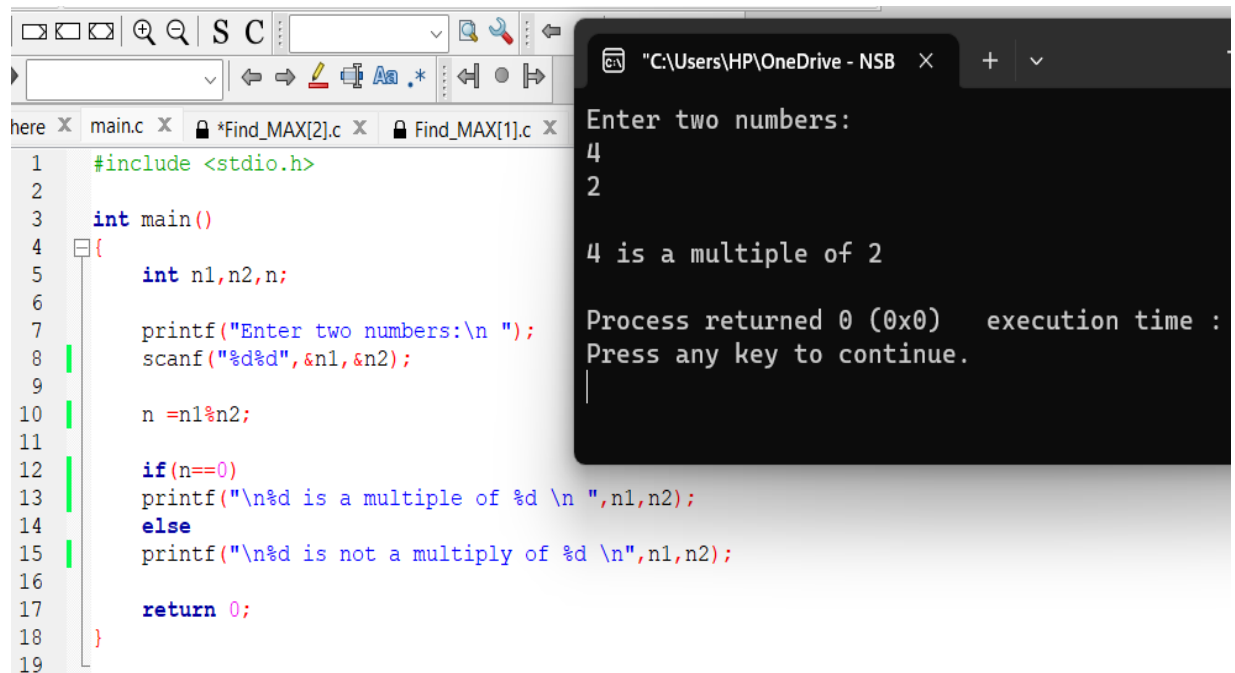
```
int main()
{
    int n1,n2,n;

    printf("Enter two numbers:\n ");
    scanf("%d%d",&n1,&n2);

    n =n1%n2;

    if(n==0)
    printf("%d is a multiple of %d ",n1,n2);
    else
    printf("%d is not a multiply of %d",n1,n2);

    return 0;
}
```



The screenshot shows a Windows environment with a code editor and a command prompt. The code editor displays the following C program:

```
1  #include <stdio.h>
2
3  int main()
4  {
5      int n1,n2,n;
6
7      printf("Enter two numbers:\n ");
8      scanf("%d%d",&n1,&n2);
9
10     n =n1%n2;
11
12     if(n==0)
13     printf("\n%d is a multiple of %d \n ",n1,n2);
14     else
15     printf("\n%d is not a multiply of %d \n",n1,n2);
16
17     return 0;
18 }
19
```

The command prompt window shows the execution of the program:

```
"C:\Users\HP\OneDrive - NSB" x + v
Enter two numbers:
4
2

4 is a multiple of 2

Process returned 0 (0x0)   execution time :
Press any key to continue.
```

```

6. #include <stdio.h>
    int main()
    {
        char ch;
        printf("Enter a character among these:
A,B,C,a,b,c,0,1,2,$,*,+, / \n ");
        scanf("%c",&ch);

        if ('A'==ch)
            printf("The integer value of A is %d\n",'A');
        else if('B'==ch)
            printf("The integer value of B is %d\n",'B');
        else if('C'==ch)
            printf("The integer value of C is %d\n",'C');
        else if('a'==ch)
            printf("The integer value of a is %d\n",'a');
        else if('b'==ch)
            printf("The integer value of b is %d\n",'b');
        else if('c'==ch)
            printf("The integer value of c is %d\n",'c');
        else if('0'==ch)
            printf("The integer value of 0 is %d\n",'0');
        else if('1'==ch)
            printf("The integer value of 1 is %d\n",'1');
        else if('2'==ch)
            printf("The integer value of 2 is %d\n",'2');
        else if('$'==ch)
            printf("The integer value of $ is %d\n",'$');
        else if('*'==ch)
            printf("The integer value of * is %d\n",'*');
        else if('+'==ch)
            printf("The integer value of + is %d\n",'+');
        else if('/'==ch)
            printf("The integer value of / is %d\n",'/');

        return 0;
    }

```

```
main.c x Find_MAX[1].c x
#include <stdio.h>
int main()
{
    char ch;
    printf("Enter a character among these: A,B,C,a,b,c,0,1,2,$,*,+/, \n ");
    scanf("%c",&ch);

    if ('A'==ch)
        printf("The integer value of A is %d\n",'A');
    else if ('B'==ch)
        printf("The integer value of B is %d\n",'B');
    else if ('C'==ch)
        printf("The integer value of C is %d\n",'C');
    else if ('a'==ch)
        printf("The integer value of a is %d\n",'a');
    else if ('b'==ch)
        printf("The integer value of b is %d\n",'b');
    else if ('c'==ch)
        printf("The integer value of c is %d\n",'c');
    else if ('0'==ch)
        printf("The integer value of 0 is %d\n",'0');
    else if ('1'==ch)
        printf("The integer value of 1 is %d\n",'1');
    else if ('2'==ch)
        printf("The integer value of 2 is %d\n",'2');
    else if ('$'==ch)
        printf("The integer value of $ is %d\n",'$');
    else if ('*'==ch)
        printf("The integer value of * is %d\n",'*');
    else if ('+'==ch)
        printf("The integer value of + is %d\n",'+');
    else if ('/'==ch)
        printf("The integer value of / is %d\n",'/');

    return 0;
}
```

"C:\Users\HP\OneDrive - NSB x + v

The integer value of A is 65

Process returned 0 (0x0) exec
Press any key to continue.

```
main.c x Find_MAX[1].c x
#include <stdio.h>
int main()
{
    char ch;
    printf("Enter a character among these: A,B,C,a,b,c,0,1,2,$,*,+/, \n ");
    scanf("%c",&ch);

    if ('A'==ch)
        printf("The integer value of A is %d\n",'A');
    else if ('B'==ch)
        printf("The integer value of B is %d\n",'B');
    else if ('C'==ch)
        printf("The integer value of C is %d\n",'C');
    else if ('a'==ch)
        printf("The integer value of a is %d\n",'a');
    else if ('b'==ch)
        printf("The integer value of b is %d\n",'b');
    else if ('c'==ch)
        printf("The integer value of c is %d\n",'c');
    else if ('0'==ch)
        printf("The integer value of 0 is %d\n",'0');
    else if ('1'==ch)
        printf("The integer value of 1 is %d\n",'1');
    else if ('2'==ch)
        printf("The integer value of 2 is %d\n",'2');
    else if ('$'==ch)
        printf("The integer value of $ is %d\n",'$');
    else if ('*'==ch)
        printf("The integer value of * is %d\n",'*');
    else if ('+'==ch)
        printf("The integer value of + is %d\n",'+');
    else if ('/'==ch)
        printf("The integer value of / is %d\n",'/');

    return 0;
}
```

"C:\Users\HP\OneDrive - NSB x +

The integer value of 1 is 49

Process returned 0 (0x0) ex
Press any key to continue.

7.

```
int main()
{
    Float
    basicsalary,additionalallowance,bonus,monthllysales
    ,grossremuneration;
    int noofyears;
    char city;

    printf("Enter basic salary: ");
    scanf("%f",&basicsalary);

    printf("Enter the no of service: ");
    scanf("%d",&noofyears);

    printf("Enter the city ('c'= Colombo, 'O'= other): ");
    scanf(" %c", &city);

    printf("Enter the monthly sales: ");
    scanf("%f",&monthllysales);

    if(noofyears>5)
        additionalallowance+=0.1*basicsalary;
    if(city=='c')
        additionalallowance+=2500;

    monthllysales=0;
    if(monthllysales>=50000)
        bonus=monthllysales*0.15;
    else if(monthllysales>=25000)
        bonus=monthllysales*0.12;
    else
        bonus=monthllysales*0.10;

    grossremuneration= basicsalary+additionalallowance+bonus;
```

```

printf("Gross Monthly Remuneration:
      %2f",grossremuneration);

return 0;
}

```

The screenshot displays a C program in a code editor and its execution in a terminal window. The code calculates the Gross Monthly Remuneration based on basic salary, years of service, city, and monthly sales.

Source Code (main.c):

```

1  int main()
2  {
3      float basicsalary, additionalallowance, bonus, monthliesales, grossremuneration;
4      int noofyears;
5      char city;
6
7      printf("Enter basic salary: ");
8      scanf("%f", &basicsalary);
9
10     printf("Enter the no of service: ");
11     scanf("%d", &noofyears);
12
13     printf("Enter the city ('c'= Colombo, 'O'= other): ");
14     scanf(" %c", &city);
15
16     printf("Enter the monthly sales: ");
17     scanf("%f", &monthliesales);
18
19     if(noofyears>5)
20         additionalallowance+=0.1*basicsalary;
21     if(city=='c')
22         additionalallowance+=2500;
23
24     monthliesales=0;
25     if(monthliesales>=50000)
26         bonus=monthliesales*0.15;
27     else if(monthliesales>=25000)
28         bonus=monthliesales*0.12;
29     else
30         bonus=monthliesales*0.10;
31
32     grossremuneration= basicsalary+additionalallowance+bonus;
33
34     printf("Gross Monthly Remuneration: %2f", grossremuneration);
35     return 0;
36 }

```

Terminal Output:

```

Enter basic salary: 5000
Enter the no of service: 5
Enter the city ('c'= Colombo, 'O'= other): c
Enter the monthly sales: 50000
Gross Monthly Remuneration: 7500.000000
Process returned 0 (0x0)   execution time : 12.396 s
Press any key to continue.

```